TMA solutions

YOUR TECHNOLOGY & INNOVATION PARTNER

AI Agent Development Training

Training Roadmap

Workshop 1:
Build Your First Production Agent

Workshop 2:
Complex Agent Implementation Lab

Workshop 3:
Enterprise Agent Solutions with NATS

- Agent Architecture & Design Patterns
- Modern Agent Frameworks Overview
- Tool Integration with MCP & Guardrails
- Advanced RAG with GraphRAG & Memory Systems
- Building Multi-Agent Systems
- Production Deployment & Operations
- Agent Evaluation, Optimization & Safety

AI Agent Development Program - 12 weeks

Prerequisites: AI Development Program - Foundation & Intermediate

AI Agent Development Training Program: Courses & Workshops

<u>**Prerequisites**</u>**: AI Foundation + Intermediate Programs**

1. **Course 1**: Agent Architecture & Design Patterns
2. **Course 2**: Modern Agent Frameworks Overview
3. **Workshop 1**: Build Your First Production Agent
4. **Course 3**: Tool Integration with MCP & Guardrails
5. **Course 4**: Advanced RAG with GraphRAG & Memory Systems
6. **Workshop 2**: Complex Agent Implementation Lab
7. **Course 5**: Building Multi-Agent Systems
8. **Course 6**: Production Deployment & Operations
9. **Workshop 3 (Optional)**: Enterprise Agent Solutions with NATS
10. **Course 7**: Agent Evaluation, Optimization & Safety

1. Agent Architecture & Design Patterns

Master the foundational patterns and architectures that power modern AI agents. Learn to design autonomous, goal-oriented systems that go beyond simple chatbots, implementing sophisticated reasoning patterns like Re-Act, Chain-of-Thought, and state management for long-running agent workflows.

Course Outline

- Evolution from LLM Applications to Autonomous Agents
- Re-Act Pattern: Reasoning and Acting in Agents
- Chain-of-Thought and Tree-of-Thought Planning
- Goal-Oriented vs Task-Oriented Agent Architectures
- State Machines and Agent Lifecycle Management
- Event-Driven and Observer Patterns for Agents
- Error Handling, Retry Logic, and Fallback Strategies
- Agent Memory Patterns: Working, Episodic, Semantic

Course Outcomes

- Design agents using established architectural patterns
- Implement Re-Act and advanced reasoning frameworks
- Build robust state management for complex workflows
- Create error-resilient autonomous systems
- Architect modular, maintainable agent codebases
- Evaluate architectural trade-offs for different use cases

2. Modern Agent Frameworks Overview

Comprehensive exploration of leading agent frameworks including LangChain with LangGraph, Pydantic AI, Microsoft AutoGen, and CrewAI. Learn the strengths, use cases, and implementation patterns of each framework to select the right tool for different agent development scenarios.

Course Outline

- LangChain & LangGraph: Chains, Agents, and Stateful Workflows
- Pydantic AI: Type-Safe Agent Development and Validation
- Microsoft AutoGen: Conversational Multi-Agent Systems
- CrewAI: Role-Based Agent Teams and Collaboration
- Framework Comparison: Strengths and Trade-offs
- Streaming, Human-in-the-Loop, and State Management
- Prompt Engineering Across Different Frameworks
- Testing and Debugging Strategies for Each Framework

Course Outcomes

- Build agents using LangChain, Pydantic AI, AutoGen, and CrewAI
- Select appropriate frameworks for specific use cases
- Implement stateful workflows with LangGraph
- Create type-safe agents with Pydantic validation
- Build conversational teams with AutoGen
- Deploy role-based agents using CrewAI

3. Workshop 1: Build Your First Production Agent

Hands-on workshop building a complete agent using LangChain and LangGraph. Create a customer support agent with stateful conversations, tool integration using MCP (Model Context Protocol), and proper error handling while implementing best practices learned in previous courses.

Workshop Outline

- Project setup with LangChain and LangGraph
- Implementing Core Agent Loop with Graph Architecture
- Building Custom MCP Tools for Agent Actions
- Adding Conversation Memory with LangGraph State
- Integrating Vector Store for Knowledge Base
- Implementing Checkpoints and Recovery
- Testing Agent Behaviors with Test Suites
- Live Demo and Architecture Review

Workshop Outcomes

- Deploy a working agent with LangGraph orchestration
- Implement MCP tools for external integrations
- Build stateful conversation management
- Create comprehensive test coverage
- Handle errors and edge cases gracefully
- Present architectural decisions to peers

4. Tool Integration with MCP & Guardrails

Master the Model Context Protocol (MCP) for standardized tool integration and implement Guardrails AI for input/output validation. Learn to build secure, reliable agents that can safely interact with external systems, APIs, and databases while preventing harmful outputs and ensuring quality.

Course Outline

- MCP Fundamentals: Tools, Resources, and Prompts
- Building Custom MCP Servers and Tools
- Function Calling: OpenAI, Anthropic
- API Integration: REST, WebSocket
- Database Operations: SQL and NoSQL
- Guardrails AI: Input Validation and Output Control
- Building Custom Validators and Safety Checks
- Tool Selection Strategies and Error Handling

Course Outcomes

- Build and deploy custom MCP servers and tools
- Implement Guardrails for input/output validation
- Create secure integrations with enterprise APIs
- Design custom validators for domain-specific needs
- Handle tool failures with graceful fallbacks
- Ensure safe and reliable agent behaviors

5. Advanced RAG with GraphRAG & Memory Systems

Explore GraphRAG for knowledge graph-enhanced retrieval and advanced memory architectures. Learn to build agents that can reason over complex relationships, answer holistic questions about datasets, and maintain sophisticated memory across sessions.

Course Outline

- GraphRAG Architecture: From Documents to Knowledge Graphs
- Community Detection and Hierarchical Summarization
- Global vs Local Search Strategies in GraphRAG
- Traditional RAG vs GraphRAG Trade-offs
- Multi-tiered Memory Architectures
- Vector Databases at Scale: Pinecone, Weaviate, Qdrant
- Memory Consolidation and Context Management
- Evaluating RAG Performance and Optimization

Course Outcomes

- Implement GraphRAG for complex data understanding
- Build knowledge graphs from unstructured text
- Design hybrid RAG systems combining approaches
- Create sophisticated memory management systems
- Optimize retrieval for accuracy and efficiency
- Evaluate and improve RAG system performance

6. Workshop 2: Complex Agent Implementation Lab

Build sophisticated multi-capability agents combining MCP tools, GraphRAG, and Guardrails. Teams create agents for real scenarios like data analysis or research automation, implementing self-correction, quality checks, and advanced reasoning with proper safety measures.

Workshop Outline

- Building a Data Analysis Agent with SQL and Visualization
- Implementing GraphRAG for Complex Document Analysis
- Adding MCP Tools for External Service Integration
- Implementing Guardrails for Safety and Quality
- Self-Reflection and Error Correction Mechanisms
- Performance Profiling and Optimization
- A/B Testing Different RAG Strategies
- Team Presentations and Code Reviews

Workshop Outcomes

- Build agents combining 5+ different capabilities
- Implement GraphRAG for knowledge-intensive tasks
- Deploy comprehensive safety guardrails
- Create self-correcting agent behaviors
- Compare different implementation strategies
- Document architectural decisions and trade-offs

7. Building Multi-Agent Systems

Master multi-agent orchestration using LangChain and LangGraph with both A2A Protocol and Event Bus architectures. Learn to build supervisor patterns, hierarchical teams, and routing based on reasoning, implementing different communication paradigms for enterprise-scale agent systems.

Course Outline

- LangGraph Multi-Agent Architectures: Supervisor and Teams
- A2A Protocol: Agent Discovery and Communication
- Event Bus Architecture: Pub/Sub for Agent Communication
- Comparing A2A vs Event Bus: Trade-offs and Use Cases
- Building Hierarchical Agent Teams with LangGraph
- Agentic Routing and Dynamic Agent Selection
- State Sharing and Coordination Between Agents
- Monitoring and Debugging Multi-Agent Workflows

Course Outcomes

- Build multi-agent systems using LangGraph orchestration
- Implement both A2A and Event Bus communication patterns
- Create supervisor-based agent architectures
- Design hierarchical teams with specialized agents
- Implement dynamic routing based on agent reasoning
- Scale LangGraph multi-agent systems for production

8. Production Deployment & Operations

Learn to deploy, scale, and maintain AI agents in production environments. Cover infrastructure, security with Guardrails, monitoring, cost optimization, and compliance requirements for running agents at scale in enterprise cloud environments.

Course Outline

- Containerization: Docker and Kubernetes for Agents
- API Design for Agent Services: REST and WebSocket
- Security: OAuth, Rate Limiting, Input Validation
- Production Guardrails: Safety at Scale
- Observability: Logging, Metrics, Tracing
- Cost Management: Token Usage and Caching
- CI/CD Pipelines for Agent Applications
- Compliance and Audit Requirements

Course Outcomes

- Deploy agents to Docker and Kubernetes
- Implement production-grade security measures
- Set up comprehensive monitoring systems
- Deploy Guardrails for production safety
- Optimize costs while maintaining performance
- Build CI/CD pipelines with automated testing

9. Workshop 3: Enterprise Agent Solutions with NATS

Design and implement enterprise-grade agent solutions using NATS messaging for high-performance agent communication. Build agents that leverage MCP for tools, NATS for scalable messaging, GraphRAG for knowledge, and Guardrails for safety, integrating with existing enterprise systems at scale.

Workshop Outline

- Enterprise Architecture with NATS Messaging System
- Building Workflow Automation with NATS Pub/Sub
- Implementing GraphRAG for Enterprise Knowledge
- Multi-Agent Collaboration via NATS JetStream
- Enterprise System Integration (SAP, Salesforce)
- Production Guardrails and Compliance
- High Availability with NATS Clustering
- Executive Presentation and ROI Analysis

Workshop Outcomes

- Deploy enterprise solution with NATS messaging
- Implement scalable pub/sub agent communication
- Integrate GraphRAG for complex data analysis
- Build resilient systems with NATS clustering
- Implement comprehensive Guardrails for safety
- Present complete solution to stakeholders

10. Agent Evaluation, Optimization & Safety

Master techniques for evaluating agent performance, implementing comprehensive safety measures with Guardrails, and optimizing agent systems. Learn to build evaluation frameworks, conduct systematic testing, and ensure responsible AI deployment with proper safety controls.

Course Outline

- Evaluation Metrics: Accuracy, Latency, Cost, Safety
- Building Comprehensive Test Suites
- Guardrails for Production: Safety and Quality
- Red Teaming and Adversarial Testing
- A/B Testing Agent Strategies
- Performance Profiling and Optimization
- Continuous Monitoring and Improvement
- Responsible AI and Ethical Considerations

Course Outcomes

- Build comprehensive evaluation frameworks
- Implement multi-layer safety systems
- Conduct systematic agent testing
- Deploy production Guardrails effectively
- Optimize cost-performance trade-offs
- Establish continuous improvement processes